



Qualification Pack



Process Operator Rolling Mills

QP Code: ISC/Q0701

Version: 1.0

NSQF Level: 5

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ISC/Q0701: Process Operator Rolling Mills

Brief Job Description

This job is all about keeping different process lines functional by ensuring continuity of process. It also involves providing the actual commands for different operations as needed and monitor the operations for any deviations. It includes inspecting the material between processes to ensure alignment to specifications.

Personal Attributes

This job requires the candidate to work independently as well as in teams. Some of the key attributes includes physical fitness, not having colour blindness, having analytical skills, problem solving attitude, high concentration levels, a sharp reflex and willingness to work in a factory environment.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [ISC/N0008: Use basic health and safety practices at the workplace](#)
2. [ISC/N0096: Work effectively with others and manage team members](#)
3. [ISC/N0706: Prepares for operation of control systems](#)
4. [ISC/N0707: Performs operation of control systems](#)

Qualification Pack (QP) Parameters

Sector	Iron and Steel
Sub-Sector	Steel, Re-Rollers
Occupation	Cold Rolling Mill
Country	India
NSQF Level	5
Credits	NA
Aligned to NCO/ISCO/ISIC Code	NCO-2004/NIL



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Minimum Educational Qualification & Experience	I.T.I with 0-6 Months of experience In similar function OR I.T.I with 3-5 Years of experience In lieu of minimum qualification the incumbent should have 4-5 years of relevant work experience under an experienced Process Operator
Minimum Level of Education for Training in School	10th Class
Pre-Requisite License or Training	Basics of metallurgy Acid handling Annealing Galvanising Basics of Hydraulics, Pneumatics and Electrical controls
Minimum Job Entry Age	18 Years
Last Reviewed On	30/12/2014
Next Review Date	31/03/2022
NSQC Approval Date	18/06/2015
Version	1.0
Reference code on NQR	2015/IS/IISSSC/00297
NQR Version	1.0



Qualification Pack

ISC/N0008: Use basic health and safety practices at the workplace

Description

This OS unit is about knowledge and practices relating to health, safety and security that candidates need to use in the workplace. It covers responsibilities towards self, others, assets and the environment. It includes understanding of risks and hazards in the workplace, along with common techniques to minimize risk, deal with accidents, emergencies, etc

Elements and Performance Criteria

Health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** Use protective clothing/equipment for specific tasks and work conditions
Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks
Glasses/gloves/visors
Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator

Health and safety procedures

To be competent, the user/individual on the job must be able to:

- PC2.** State the name and location of people responsible for health and safety in the workplace
Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.
- PC3.** State the names and location of documents that refer to health and safety in the workplace
- PC4.** Identify job-site hazardous work and state possible causes of risk or accident in the workplace
Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.
- PC5.** Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role
Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors
- PC6.** State location of general health and safety equipment in the workplace



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- PC7.** Inspect for faults, set up and safely use steps and ladders in general use
Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc.
Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.
- PC8.** Work safely in and around trenches, elevated places and confined areas
- PC9.** Lift heavy objects safely using correct procedures
- PC10.** Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces
- PC11.** Identify common hazard signs displayed in various areas
- PC12.** Retrieve and/or point out documents that refer to health and safety in the workplace

Fire safety procedures

To be competent, the user/individual on the job must be able to:

- PC13.** Use the various appropriate fire extinguishers on different types of fires correctly.
Fire extinguishers: Sand Water Foam Co2 Dry powder
Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents)
Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.
- PC14.** Demonstrate rescue techniques applied during fire hazard
- PC15.** Demonstrate good housekeeping in order to prevent fire hazards
- PC16.** Demonstrate the correct use of a fire extinguisher

Emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

- PC17.** Demonstrate how to free a person from electrocution
- PC18.** Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.
- PC19.** Demonstrate basic techniques of bandaging
- PC20.** Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below :
Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.
- PC21.** Perform and organize loss minimization or rescue activity during an accident in real or simulated environments
- PC22.** Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases
- PC23.** Demonstrate the artificial respiration and the CPR Process
- PC24.** Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work



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- PC25.** Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer
- PC26.** Demonstrate correct method to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** State the names (and job titles if applicable), and describe where to find, all the people responsible for health and safety in a workplace
- KU2.** State the names and location of documents that refer to health and safety in the workplace
- KU3.** Meaning of hazards and risks
- KU4.** Health and safety hazards commonly present in the work environment and related precautions
- KU5.** Possible causes of risk, hazard or accident in the workplace and why risk and/or accidents are possible
- KU6.** Activities and causes of risk and accident
- KU7.** Methods of accident prevention
- KU8.** Safe working practices when working with tools and machines
- KU9.** Safe working practices while working at various hazardous sites
- KU10.** Where to find all the general health and safety equipment in the workplace
- KU11.** Various dangers associated with the use of electrical equipment
- KU12.** Preventative and remedial actions to be taken in the case of exposure to toxic materials. Exposure: ingested, contact with skin, inhaled Preventative action: ventilation, masks, protective clothing/equipment Remedial action: immediate first aid, report to supervisor Materials: solvents, flux, lead
- KU13.** Importance of using protective clothing/equipment while working
- KU14.** Precautionary activities to prevent the fire accident Activities and causes: Physical actions Reading Listening to and giving instructions Inattention Sickness and incapacity (e.g. Drunkenness) Health hazards (e.g. Untreated injuries and contagious illness)
- KU15.** Various causes of fire
- KU16.** Techniques of using the different fire extinguishers
- KU17.** Different methods of extinguishing fire
- KU18.** Rescue techniques applied during a fire hazard
- KU19.** Various types of safety signs and what they mean
- KU20.** Appropriate basic first aid treatment relevant to the condition e.g. Shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries
- KU21.** Content of written accident report



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- KU22.** Potential injuries and ill health associated with incorrect manual handling
- KU23.** Safe lifting and carrying practices
- KU24.** Personal safety, health and dignity issues relating to the movement of a person by others
- KU25.** Potential impact to a person who is moved incorrectly

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Read and comprehend basic content to read labels, charts, signages
- GS2.** Read and comprehend basic English to read manuals of operations
- GS3.** Read and write an accident/incident report in local language or English
- GS4.** Question co-workers appropriately in order to clarify instructions and other issues
- GS5.** Give clear instructions to co-workers, subordinates others
- GS6.** Make appropriate decisions pertaining to the concerned area of work with respect to intended work objective, span of authority, responsibility, laid down procedure and guidelines
- GS7.** Plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS8.** Remain congenial while discussing and debating issues with co-workers
- GS9.** Follow appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS10.** Ask for, provide and receive required assistance where possible to ensure achievement of work related objectives
- GS11.** Thank co-workers for any assistance received
- GS12.** Offer appropriate respect based on mutuality and respect for fellow workmanship and authority
- GS13.** Think through the problem, evaluate the possible solution(s) and suggest an optimum /best possible solution(s)
- GS14.** Identify immediate or temporary solutions to resolve delays
- GS15.** Identify sources of support that can be availed of for problem solving for various kind of problems
- GS16.** Report problems that you cannot resolve to appropriate authority
- GS17.** Identify cause and effect relations in their area of work
- GS18.** Use cause and effect relations to anticipate potential problems and their solution



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Health and safety</i>	5	5	-	-
PC1. Use protective clothing/equipment for specific tasks and work conditions Protective clothing includes: Leather or asbestos gloves Flame proof aprons Flame proof overalls buttoned to neck Cuff less (without folds) trousers Reinforced footwear Helmets/hard hats Cap and shoulder covers Ear defenders/plugs Safety boots Knee pads Particle masks Glasses/gloves/visors Equipment includes: Hand shields Machine guards Residual current devices Shields Dust sheets Respirator	5	5	-	-
<i>Health and safety procedures</i>	15	40	-	-
PC2. State the name and location of people responsible for health and safety in the workplace Various areas are listed below: On chemical containers Equipment Packages Inside buildings Open areas and public spaces, etc.	-	4	-	-
PC3. State the names and location of documents that refer to health and safety in the workplace	-	1	-	-
PC4. Identify job-site hazardous work and state possible causes of risk or accident in the workplace Hazards include: Working with electrical and thermal tools and equipment Sharp edged and heavy tools Heated metals Oxyfuel and gas cylinders Welding radiation Surfaces: sharp, slippery, uneven, chipped, broken, etc. Substances: chemicals, gas, oxy-fuel, fumes, dust, etc. Physical: working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages, etc. Electrical: power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC5. Carry out safe working practices while dealing with hazards to ensure the safety of self and others state methods of accident prevention in the work environment of the job role Safe working practices include: Using protective clothing and equipment Putting up and reading safety signs Handle tools in the correct manner and store and maintain them properly Keep work area clear of clutter, spillage and unsafe object lying casually While working with electricity take all electrical precautions like insulated clothing, adequate equipment insulation, use of control equipment, dry work area, switch off the power supply when not required, etc. Safe lifting and carrying practices Use equipment that is working properly and is well maintained Take due measures for safety while working in confined places, trenches or at heights, etc. Including safety harness, fall arrestors, etc. Methods are: Training in health and safety procedures Using health and safety procedures Use of equipment and working practices (such as safe carrying procedures) Safety notices, advice Instruction from colleagues and supervisors	5	5	-	-
PC6. State location of general health and safety equipment in the workplace	-	5	-	-
PC7. Inspect for faults, set up and safely use steps and ladders in general use Faults : Corrosion of metal components Deterioration Splits and cracks timber components Imbalance Loose rungs Nuts or bolts, etc. Set up: Firm/level base Clip/lash down Leaning at the correct angle, etc.	-	5	-	-
PC8. Work safely in and around trenches, elevated places and confined areas	-	4	-	-
PC9. Lift heavy objects safely using correct procedures	-	4	-	-
PC10. Apply good housekeeping practices at all times. Good housekeeping practices: Clean/tidy work areas Removal/disposal of waste products Protect surfaces	-	1	-	-
PC11. Identify common hazard signs displayed in various areas	5	1	-	-
PC12. Retrieve and/or point out documents that refer to health and safety in the workplace	-	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Fire safety procedures</i>	10	15	-	-
PC13. Use the various appropriate fire extinguishers on different types of fires correctly. Fire extinguishers: Sand Water Foam Co2 Dry powder Fires: Class A: Ordinary solid combustibles, e.g. wood, paper, cloth, plastic, charcoal etc. Class B: Flammable liquids and gases, e.g. gasoline, propane, diesel fuel, tar, cooking oil and similar substances Class C: Electrical equipment e.g. appliances, wiring, breaker panels etc. (these categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity) Class D: Combustible metals such as magnesium, titanium, and sodium (these fires burn at extremely high temperatures and require special suppression agents) Causes of fires: Heating of metal Spontaneous ignition Sparking, Electrical heating Loose fires (e.g. Smoking, welding, etc.) Chemical fires, etc.	5	5	-	-
PC14. Demonstrate rescue techniques applied during fire hazard	5	5	-	-
PC15. Demonstrate good housekeeping in order to prevent fire hazards	-	1	-	-
PC16. Demonstrate the correct use of a fire extinguisher	-	4	-	-
<i>Emergencies, rescue and first-aid procedures</i>	15	45	-	-
PC17. Demonstrate how to free a person from electrocution	-	5	-	-
PC18. Administer appropriate first aid to victims as required e.g. in case of bleeding, burns, choking, electric shock, poisoning etc.	5	5	-	-
PC19. Demonstrate basic techniques of bandaging	-	5	-	-
PC20. Respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments. few General health and safety equipment are mentioned below : Fire extinguishers First aid equipment Safety instruments and clothing Safety installations, e.g. Fire exits, exhaust fans etc.	5	5	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC21. Perform and organize loss minimization or rescue activity during an accident in real or simulated environments	-	5	-	-
PC22. Administer first aid to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	-	5	-	-
PC23. Demonstrate the artificial respiration and the CPR Process	-	5	-	-
PC24. Participate in emergency procedures. Emergency procedures are: Raising alarm Safe/efficient evacuation Correct means of escape Correct assembly point Roll call Correct return to work	-	4	-	-
PC25. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report should capture: Name Date/time of incident Date/time of report, Location Environment conditions Persons involved Sequence of events Injuries sustained Damage sustained Actions taken Witnesses Supervisor/manager notified Documents: Fire notices Accident reports Safety instructions for equipment and procedures Company notices and documents Legal documents (e.g. Government notices) Job titles: ISC/N0008: Use basic health and safety practices at the workplace Health and safety officer First aid officer Fire officer	5	5	-	-
PC26. Demonstrate correct method to move injured people and others during an emergency	-	1	-	-
NOS Total	45	105	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0008
NOS Name	Use basic health and safety practices at the workplace
Sector	Iron and Steel
Sub-Sector	Steel, Sponge Iron, Ferro Alloys, Re-Rollers, Refractory
Occupation	Mechanical Maintenance
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025



Qualification Pack

ISC/N0096: Work effectively with others and manage team members

Description

This unit covers basic etiquett and competencies that a candidate is required to possess and demonstrate in their behaviour and interactions with others at the workplace. These cover areas such as communication etiquette, discipline, listening, handling conflict and grievances

Scope

This unit/task covers the following:

- Communicate effectively with team members, co-workers and supervisors
- Encourage friendly working environment
- Use appropriate titles and terms of respect
- Follow code of conduct and ethics as applicable
- Follow organisation specific policies
- Demonstrate effective behaviours for team work
- Ensure engagement of team members through on-the job handholding & support

Elements and Performance Criteria

Communicate effectively with coworkers and supervisors

To be competent, the user/individual on the job must be able to:

- PC1.** accurately receive, absorb and share information and instructions from the supervisor and fellow workers, getting clarification where required
- PC2.** display appropriate communication etiquette while working
- PC3.** display active listening skills while interacting with others at work
- PC4.** use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism

Demonstrate effective behaviours for team work

To be competent, the user/individual on the job must be able to:

- PC5.** display helpful behaviour by assisting others in performing tasks in effective manner
- PC6.** consult with and assist others to maximize effectiveness and efficiency in carrying out tasks
- PC7.** demonstrate responsible and disciplined behaviours at the workplace
- PC8.** escalate grievances and problems to superiors

Ensure engagement of team members through on-the job handholding & support

To be competent, the user/individual on the job must be able to:

- PC9.** communicate day-to-day objectives, instructions etc. to team members
- PC10.** guide the team members to manage day-to-day issues at work
- PC11.** gather concerns, feedback from team members and convey them to appropriate authorities
- PC12.** escalate grievances and problems to superiors

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** various categories of people that one is required to communicate and coordinate with in the organization
- KU6.** importance of teamwork in organizational and individual success
- KU7.** various components of effective communication and importance of effective communication while communicating professional circles
- KU8.** key elements of active listening and importance of active listening
- KU9.** importance of ethics for professional success
- KU10.** importance of discipline and disciplined behaviour for professional success
- KU11.** common reasons, importance and ways of managing interpersonal conflict effectively
- KU12.** expressing and addressing grievances appropriately and effectively

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read and comprehend basic content to read instructions, labels, charts, signages
- GS2.** write notes in log books etc.
- GS3.** prepare incident reports, as appropriate
- GS4.** question co-workers appropriately in order to clarify instructions and other issues
- GS5.** communicate with co-workers
- GS6.** share information with supervisors
- GS7.** make appropriate decisions pertaining to the concerned area of work with respect to safely achieving work objective following the laid down procedure and guidelines
- GS8.** plan and organize their own work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS9.** identify sources of support that can be availed of for problem solving for various kind of problems
- GS10.** seek appropriate assistance from other sources to resolve problems
- GS11.** report problems that cannot be resolved to appropriate authority
- GS12.** identify cause and effect relations in own area of work
- GS13.** use cause and effect relations to anticipate potential problems and their solution



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate effectively with coworkers and supervisors</i>	-	30	-	-
PC1. accurately receive, absorb and share information and instructions from the supervisor and fellow workers, getting clarification where required	-	5	-	-
PC2. display appropriate communication etiquette while working	-	10	-	-
PC3. display active listening skills while interacting with others at work	-	10	-	-
PC4. use appropriate tone, pitch and language to convey politeness, assertiveness, care and professionalism	-	5	-	-
<i>Demonstrate effective behaviours for team work</i>	-	30	-	-
PC5. display helpful behaviour by assisting others in performing tasks in effective manner	-	10	-	-
PC6. consult with and assist others to maximize effectiveness and efficiency in carrying out tasks	-	10	-	-
PC7. demonstrate responsible and disciplined behaviours at the workplace	-	5	-	-
PC8. escalate grievances and problems to superiors	-	5	-	-
<i>Ensure engagement of team members through on-the-job handholding & support</i>	10	30	-	-
PC9. communicate day-to-day objectives, instructions etc. to team members	5	10	-	-
PC10. guide the team members to manage day-to-day issues at work	-	5	-	-
PC11. gather concerns, feedback from team members and convey them to appropriate authorities	5	10	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. escalate grievances and problems to superiors	-	5	-	-
NOS Total	10	90	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0096
NOS Name	Work effectively with others and manage team members
Sector	Iron and Steel
Sub-Sector	Steel
Occupation	Coke Making
NSQF Level	4
Credits	TBD
Version	1.0
Last Reviewed Date	30/12/2014
Next Review Date	31/03/2022
NSQC Clearance Date	20/07/2015



Qualification Pack

ISC/N0706: Prepares for operation of control systems

Description

This unit is about inspecting the incoming coil according to specifications and prepare control system for carrying out the operation at different work stations.

Elements and Performance Criteria

Inspect the incoming coil according to specifications and understand the given job

To be competent, the user/individual on the job must be able to:

- PC1.** read and understand the specifications of the job to be carried out
- PC2.** inspect the incoming coil for dimensions, telescopicity, quality/ physical defect according to the process chart and ensure they tally with pdi (primary data input)
- PC3.** take necessary actions to get any deviations from specifications corrected
- PC4.** report any issues that may be out of his/her scope to supervisor

Load the coil for operation

To be competent, the user/individual on the job must be able to:

- PC5.** understand how to load the coil on the rolling system correctly
- PC6.** load or stack the coil(s) for conducting the operation, as appropriate
- PC7.** arrange the coil as per diameter and width to ensure correct placement
- PC8.** maintain the continuity of operation by welding the forward coil with the trailing coil, where appropriate

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job specific documents e.g. daily work checklist and importance of the same
- KU2.** safety policy of the company, as applicable to his area of working
- KU3.** emergency plan of the specific work site
- KU4.** risk and impact of not following defined safe working practice/work instructions, or spi (standard practice instructions)
- KU5.** escalation matrix for reporting identified problems
- KU6.** understand the physical properties and the features of coils, the effect of temperature, time, line-speed, burner flames, molten zinc, acid concentration, bath characteristics, alkali solution etc. on the coils as applicable in the different processing lines.
- KU7.** read and interpret process charts, control plans, job manuals, safety and other instruction documents etc.
- KU8.** understand how the pickling/ rolling/annealing/ electrolytic cleaning/galvanising systems work and the essential associated parts/machines for the specific process
- KU9.** use of different interlocks and operation sequence

Generic Skills (GS)



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User/individual on the job needs to know how to:

- GS1.** read equipment manuals and process documents given by the equipment supplier to understand the equipment and processes better
- GS2.** make notes on any deviation from desirable specifications/characteristics of the interim processes/work stations and inform supervisor, as required
- GS3.** answer the queries raised by the operative team as well as intra-company departments
- GS4.** discuss task lists, schedules, and work-loads with team members
- GS5.** prioritize and execute tasks within scheduled time limits
- GS6.** avoid absenteeism and maintain quality of work
- GS7.** work in a disciplined environment and adhere to working norms of the organisation
- GS8.** be punctual and adhere to timelines



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Inspect the incoming coil according to specifications and understand the given job</i>	15	70	-	-
PC1. read and understand the specifications of the job to be carried out	5	15	-	-
PC2. inspect the incoming coil for dimensions, telescopicity, quality/ physical defect according to the process chart and ensure they tally with pdi (primary data input)	5	25	-	-
PC3. take necessary actions to get any deviations from specifications corrected	5	15	-	-
PC4. report any issues that may be out of his/her scope to supervisor	-	15	-	-
<i>Load the coil for operation</i>	15	100	-	-
PC5. understand how to load the coil on the rolling system correctly	5	20	-	-
PC6. load or stack the coil(s) for conducting the operation, as appropriate	-	20	-	-
PC7. arrange the coil as per diameter and width to ensure correct placement	5	20	-	-
PC8. maintain the continuity of operation by welding the forward coil with the trailing coil, where appropriate	5	40	-	-
NOS Total	30	170	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0706
NOS Name	Prepares for operation of control systems
Sector	Iron and Steel
Sub-Sector	Steel, Re-Rollers
Occupation	Cold Rolling Mill Operations
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	30/12/2014
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015



Qualification Pack

ISC/N0707: Performs operation of control systems

Description

This unit is about operating the control system to give appropriate commands to carry out the operation, checking that the strip matches with the required specifications and then prepare the strip for the next process

Elements and Performance Criteria

Check the strip matches specifications

To be competent, the user/individual on the job must be able to:

- PC1.** check and compare the specifications of the coil with those in system/record and ensure correctness of the requirement

Give appropriate commands for operation

To be competent, the user/individual on the job must be able to:

- PC2.** set appropriate parameters (like pressure, current, speed, time-temperature cycle, fillage, concentration, tension etc.) as applicable for welding / pickling/ rolling/ annealing/ electrolytic cleaning/galvanising according to coil specifications and give command for carrying out the operation
- PC3.** monitor the process parameters during operation e.g. rpm, temperature, line-tension, pressure, concentration, line-speed, fillage, coating thickness, etc.
- PC4.** give necessary commands for the next operation (coil cooling, skin-passing, drying, side trimming, chromating, logo printing, oiling etc. as appropriate)

Prepare and dispatch for next operation

To be competent, the user/individual on the job must be able to:

- PC5.** unload the coil, recoil, weigh, strap, label and dispatch to the next station
- PC6.** report any issues to the supervisor that may be beyond his/her scope

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** job specific documents e.g. specification documents of parameters and importance of the same
- KU2.** risk and impact of not following defined procedures/work instructions
- KU3.** escalation matrix for reporting identified problems
- KU4.** implications of delays in process to the company
- KU5.** company safety rules, as applicable to his area
- KU6.** read control parameters on computer screen & take appropriate operating decision accordingly
- KU7.** operate desktop computer & be conversant with the latest software in use
- KU8.** knowledge of rolling, pickling, annealing, galvanizing etc. processes along with that of basic metallurgy to understand deeper issues regarding material behaviour



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- KU9.** working knowledge of the entry, process and exit parameters of the continuous process & be familiar with the relevant parameters of any batch process involved
- KU10.** identify different parameters & defects to be segregated and noted before passing material e.g. under/over pickling, telescopicity, stickers, dross pick up, surface defects etc.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** construct simple sentences & write technical observations, memos & requisitions in english, hindi and local language (e/h/l)
- GS2.** read equipment manuals and process documents given by the equipment supplier to understand the equipment and processes better
- GS3.** read, understand, follow/apply memo, reports, instruction manuals, quality control charts and safety documents
- GS4.** communicate with supervisors/juniors/colleagues/casual workers in english/hindi/local language (e/h/l)
- GS5.** respond objectively & orally to questions asked of him in e/h/l
- GS6.** discuss task lists, schedules, and work-loads with the team members
- GS7.** prioritize and execute tasks within scheduled time limits
- GS8.** avoid absenteeism and maintain quality of work
- GS9.** work in a disciplined environment and adhere to working norms of the organisation
- GS10.** be punctual and adhere to timelines



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Check the strip matches specifications</i>	20	50	-	-
PC1. check and compare the specifications of the coil with those in system/record and ensure correctness of the requirement	20	50	-	-
<i>Give appropriate commands for operation</i>	80	350	-	-
PC2. set appropriate parameters (like pressure, current, speed, time-temperature cycle, fillage, concentration, tension etc.) as applicable for welding / pickling/ rolling/ annealing/ electrolytic cleaning/galvanising according to coil specifications and give command for carrying out the operation	50	200	-	-
PC3. monitor the process parameters during operation e.g. rpm, temperature, line-tension, pressure, concentration, line-speed, fillage, coating thickness, etc.	20	100	-	-
PC4. give necessary commands for the next operation (coil cooling, skin-passing, drying, side trimming, chromating, logo printing, oiling etc. as appropriate)	10	50	-	-
<i>Prepare and dispatch for next operation</i>	5	45	-	-
PC5. unload the coil, recoil, weigh, strap, label and dispatch to the next station	-	30	-	-
PC6. report any issues to the supervisor that may be beyond his/her scope	5	15	-	-
NOS Total	105	445	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	ISC/N0707
NOS Name	Performs operation of control systems
Sector	Iron and Steel
Sub-Sector	Steel, Re-Rollers
Occupation	Cold Rolling Mill Operations
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	30/12/2014
Next Review Date	31/03/2022
NSQC Clearance Date	18/06/2015

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.



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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
ISC/N0008.Use basic health and safety practices at the workplace	45	105	-	-	150	15
ISC/N0096.Work effectively with others and manage team members	10	90	-	-	100	10
ISC/N0706.Prepare for operation of control systems	30	170	-	-	200	20
ISC/N0707.Performs operation of control systems	105	445	-	-	550	55
Total	190	810	-	-	1000	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
5S	Technique of maintaining orderliness “Japanese terminology.
CP	Control Plan.
WI	Work Instructions.



Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.